DEEBAK M



deebakarbeiten@gmail .com | linkedin.com/in/deebak-m-a9567b30a | +91 8489692463

CAREER OBJECTIVE

Final-year Chemical Engineering student seeking an entry-level Process,Production Engineer role to apply knowledge of process design, material and energy balances, unit operations, process simulation, equipment fundamentals, and process safety. Eager to contribute to safe, efficient, and optimized plant operations while developing practical industrial expertise.

EDUCATION

- **B.Tech – Chemical Engineering (7.9 CGPA)**
  Erode Sengunthar Engineering College, Thudupathi, Perundurai (2023-27)
- **HigherSecondary(ClassXII) (7 1.1%)**
  Municipal Higher Secondary School, Padmavathipuram (2022-23)
- **SecondarySchool(ClassX)**
  Government Higher Secondary School, Lalgudi

INTERNSHIP

- **Company: Kothari Sugars and Chemicals Limited, Kattur, Tamil Nadu**
  Completed industrial training in chemical process operations Gained exposure to sugar manufacturing processes and plant operations
  Observed unit operations and process flow in a large-scale industry
  Developed understanding of safety practices and industrial standards
- **Company: Goflex Wire and Cables (Unit of CRI Pumps),Tiruppur, Tamil Nadu**
  Observed cable manufacturing and production processes.Gained exposure to production methods and workflow in a manufacturing environment.Learned about industrial safety standards, process safety practices, and workplace safety procedures.Understood quality control and standard operating procedures (SOPs).Worked with the production team to understand day-to-day plant operations.

PROJECTS

- **Water pH and Temperature Monitoring System (Mini Project)**
  Developed a real-time water quality monitoring system using pH and temperature sensors..Monitored and displayed water quality parameters for analysis..Gained practical experience in sensor integration and process monitoring.
- **Effect of Flow Arrangements in Heat Exchanger Performance Analysis (Mini Project)**
  Developed a MATLAB program to analyze heat exchanger performance.Compared different flow arrangements with a focus on counter-flow heat exchangers.Evaluated heat transfer effectiveness and temperature profiles.Gained practical experience in heat transfer analysis and MATLAB programming.
- **Green Hydrogen Production Process Design using DWSIM (Major Project | Ongoing)**
  Developed a Process Flow Diagram (PFD) for green hydrogen production.Performed material and energy balance calculations.Simulated the process using DWSIM.Selected and sized key process equipment.Focused on process optimization and sustainable hydrogen production.

TECHNICAL SKILLS

- Unit operations,PFD & P&ID Basics,Process safety Fundamentals
  Software tools : DWSIM,Aspen plus,MS Excel,MATLAB & Python (Basic computaions,Programing),AutoCAD plant 3D,HTRI softwares basic knowledge
- Safety knowledges in 5S Methodology,6M Analysis,Basic Safety Measures
- Strong foundation in core chemical engineering subjects

KEY COMPETENCIES

Process understanding | **Team Work** | **Safety Awareness** | **Communication**

Logical Thinking | **Attention to Detail** | **Time Management**

WORKSHOP & CERTIFICATIONS

- Attended Alchemy'25 Workshop at National Institute of Technology Tiruchirappalli (Topics: Fundamentals of Aspen Optimization, MATLAB, Data Analytics)
- Participated in workshop on "Current Trends in Chemistry" at Kongu Engineering College
- Attended workshop "Design the Future: Tools to Transform Ideas into Solutions" at KPR Institute of Engineering and Technology
- Completed workshop on "Fundamentals of CNC/VMC" at Erode Sengunthar Engineering College
- Presented paper on "Green Chemistry" at Hindustan College of Engineering and Technology
- Presented paper on "Smart Water Purification" at Excel Engineering College
- Waste to Energy Conversion, NPTEL (via Indian Institute of Technology Roorkee) | Duration: Jan – Mar 2026 (8 weeks) | Score: 73%
- Basic knowledge in Advance Process Heat Exchanger Design with HTRI
- Professional Certification Course on Process Safety and Risk Analysis – IIChE Training Institute, Kolkata (In Progress, Expected Completion: August 2026)
- MATLAB Onramp — MathWorks | Completed: August 2026 | Learned MATLAB fundamentals, including variables, arrays, basic programming, and data visualization

ADDITIONAL INFORMATION

- Languages :
  Tamil – Native Proficiency
  English – Professional Working Proficiency
  German – A2 Level
  Hindi – Basic (Prathmic Level)
- Interests :
  Developing my technical skills
  Understanding abroad work culture and life style
  Sketch and art works